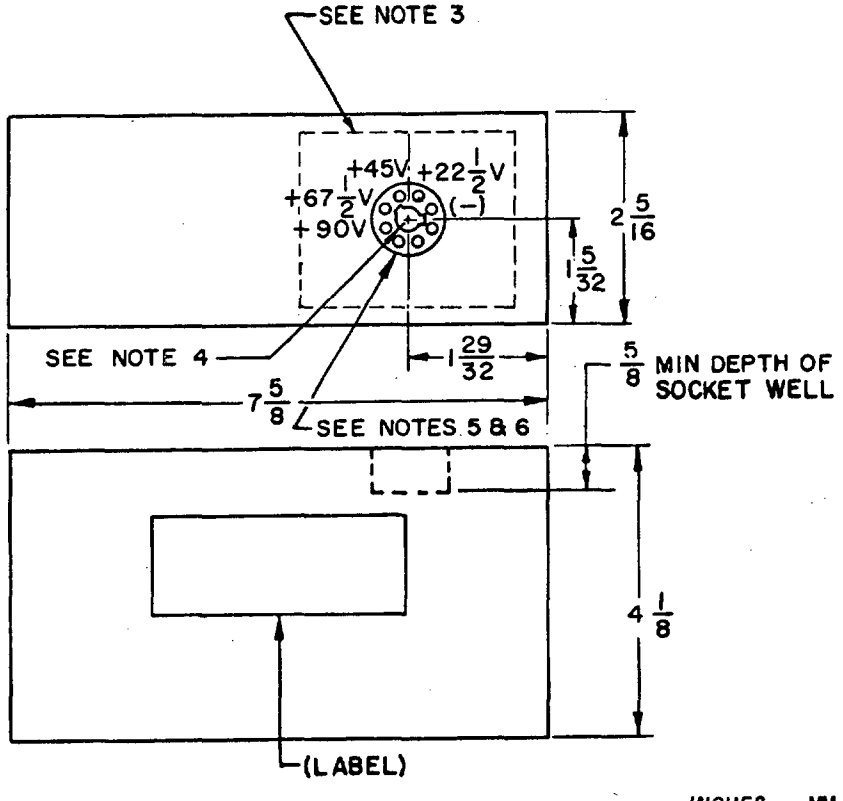
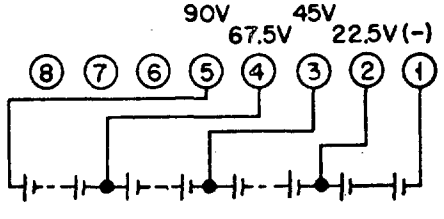
MILITARY SPECIFICATION SHEET

BATTERY, DRY, BA-419/U

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the dry battery described herein shall consist of this document and the latest issue of Specification MIL-B-18.




TYPICAL SCHEMATIC

| INCHES | MM |
|---|---|
| 1/16 | 1.59 |
| 5/8 | 15.88 |
| 1 1/16 | 26.99 |
| 1 5/32 | 29.37 |
| 1 29/32 | 48.42 |
| 2 5/16 | 58.74 |
| 4 1/8 | 104.78 |
| 7 5/8 | 193.68 |

NOTES:

1. All dimensions shown on figure are in inches. Unless otherwise specified, tolerance is ±1/16.

(D) denotes changes



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NOTES: (contd)

2. Metric equivalents specified in table are for general information only and are based upon 1 inch = 25.4 millimeters.

3. Size of insulating plate at base of socket is optional.

4. Pin-circle center.

5. The jacket opening shall be 1 1/16 minimum diameter, and shall be positioned so that its center is within a 1/16 diameter circle whose center coincides with the pin circle center.

6. Socket type B is required. (See general specification for details.)

REQUIREMENTS:

Dimensions and configuration: See figure.

Nominal voltage: 90 volts.
Tapped at: 67.5 volts.
Tapped at: 45 volts .
Tapped at: 22.5 volts .

Usual number and type of cells: 60 "A" cells.

Usual cell connection: Series.

Terminals: 8-hole socket.

Weight (Maximum) : 4 pounds, 1 ounce .

(D) Capacity tests: When the battery is tested in accordance with the methods of examination and test of this Specification, the minimum capacity-test requirements shall be not less than the minimum time specified for SLD or SLT.

(D) First Article inspection:

Visual and mechanical (External).
Battery voltage.
Vibration test.
Mechanical-shock test per MIL-STD-202, method 213, test condition I.
Insulation resistance test per MIL-STD-202, method 302, test condition Letter B, tolerance ±20 volts.
Capacity, D (without storage).
Jacket integrity test.

(D) Quality conformance inspection:

Visual and mechanical (External).
Battery voltage.
Insulation-resistance test per MIL-STD-202, method 302, test condition Letter B, tolerance ±20 volts.
Vibration test.
Mechanical-shock test per MIL-STD-202, method 213, test condition I.
Jacket integrity test.
Electrode leakage.
Capacity, D & T.



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Methods of examination and test:

Capacity tests: See requirements for capacity specified herein.

| Storage - | Test | Period |
|---|---|---|
| | D | 15 months |
| | T | 90 days |

Discharge - The battery shall be discharged through 1,600 ohms for 2 minutes, then through 3,600 ohms for 18 minutes, for a period of 4 hours, then on open circuit for 20 hours. This cycle shall be repeated 5 days per week continuously to a test-end voltage of 66.0 volts.

Closed circuit voltage - Use TS-183( )/U test per special marking or, with minimum permissible voltage specified for TS-183( )/U, use a load resistance value of 4,879 ohms.

Special marking on each unit package:

TO TEST THIS BATTERY WITH TS-183 ( )/U:

| USE JACK NO. | MINIMUM PERMISSIBLE VOLTAGE |
|---|---|
| 27 | 84.0 |

Unless expressly authorized, this test information shall apply to Army and Air Force applications only.

Custodian:
Army - ER
Navy - SH
Air Force - 99

Preparing activity:
Army - ER

Review activities:
Army - MI

Project Number: (6135-0155-62)

User activities:
Army - ME
Navy - YD, MC



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